PMx Weekly Digest: April 26, 2026

Paper 1: Intestinal Organoid-Based Mathematical Modeling Predicts Clinical Gastrointestin

Bibliographic Information

- **Title**: Intestinal Organoid-Based Mathematical Modeling Predicts Clinical Gastrointestinal Toxicity of Oral Oncology Drugs
- **Authors**: Pin C, et al.
- **Journal**: CPT: Pharmacometrics & Systems Pharmacology, 2026
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Paper Type: POPK_PKPD

Quick Take

This proof-of-concept study integrates human small intestinal organoid (hSIO) cytotoxicity data with a physiologically-based pharmacokinetic (PBPK) platform and a dynamic human epithelial model to predict clinical diarrhea induced by oral antiproliferative drugs. The key insight is that enterocyte free drug concentration—not plasma exposure—must serve as the surrogate for crypt exposure to accurately translate in vitro toxicity to clinical GI injury. Drug discovery, clinical pharmacology, and regulatory scientists should review this work for its human-relevant, model-based approach to early safety assessment and dose selection.

Executive Summary

The authors present a Quantitative Toxicology Systems (QST) framework that bridges in vitro hSIO proliferation toxicity data to clinical gastrointestinal toxicity through a multi-scale mathematical model. By coupling drug-specific, scale-free toxicity functions derived from hSIOs with Simcyp PBPK predictions of local epithelial exposure and a dynamic model of human intestinal epithelial homeostasis, the study demonstrates that enterocyte free drug concentration is the critical surrogate for crypt exposure when predicting epithelial injury for orally administered drugs. The framework successfully discriminates toxicity profiles across mechanistically distinct classes—predicting minimal injury for ribociclib and substantial disruption for abemaciclib, and accurately ranking EGFR-TKI diarrhea risk—and introduces a human-centric paradigm for preclinical GI safety assessment that aligns with modern regulatory expectations for model-informed drug development and reduced animal use.

Scientific Context & Motivation

Gastrointestinal toxicity, particularly diarrhea, affects 50–80% of patients receiving antiproliferative cancer therapies and frequently necessitates dose reduction or treatment interruption, compromising oncologic efficacy. Current preclinical strategies lack robust quantitative methods for early GI toxicity prediction, and while human small intestinal organoids (hSIOs) recapitulate crypt physiology, translating in vitro cytotoxicity into clinically actionable dosing predictions remains a major challenge. A critical knowledge gap is the appropriate surrogate for drug expo-

sure at the site of action—proliferative cells within intestinal crypts—for oral drugs, where luminal absorption dynamics create large local concentration gradients. This study addresses these gaps by developing a QST modeling approach that integrates hSIO-derived exposure-toxicity relationships with PBPK-based local exposure estimates and a dynamic epithelial model to simulate crypt proliferation impairment and epithelial recovery, ultimately predicting clinical diarrhea incidence and severity prior to human testing.

⚡ Methodological Snapshot

The study employs a three-component translational modeling strategy. First, an in vitro hSIO experimental platform coupled with a four-compartment ordinary differential equation (ODE) model (proliferative, non-proliferative, dead, and waste compartments) is used to quantify drug-specific cytotoxicity parameters from time-course imaging data. Second, a dynamic human epithelial cell model simulates crypt-to-villus epithelial homeostasis and drug-induced perturbation, into which the scale-free hSIO toxicity functions are embedded. Third, Simcyp PBPK models (including the M-ADAM enterocyte model) generate time courses of free drug concentration in plasma and enterocytes, which serve as alternative boundary estimates for crypt exposure. The predicted epithelial injury is then linked to clinical diarrhea probability via a nonlinear multinomial logistic regression fitted to aggregated clinical trial data.

⌧⌧ Structural Model Breakdown

The hSIO cell dynamics model comprises proliferative MCM2+ cells (P), non-proliferative cells (N), dead cells (D), and accumulated cell waste (W). Proliferative cells transition to non-proliferative states with feedback regulated by P/p_{ref}, and non-proliferative cells senesce regulated by N/n_{ref}. Waste induces secondary killing via a Hill function. Drug effects are introduced through scale-free functions $F_{arrest}(Drug)$ and $F_{kill}(Drug)$ that modulate the proliferation rate and number of proliferative cells, respectively. These same functions are applied in the human epithelial model, which represents the epithelium as a two-dimensional strip from crypt base to villus tip with four physical compartments: stem cells, proliferative progenitors in the crypt, and non-proliferative cells in the crypt and villus. The model maintains homeostasis through balanced proliferation, differentiation, and shedding, and recovers after drug challenge. Villus injury is quantified as the percentage of missing villus epithelial cells.

Detailed Methodological Analysis

Modeling Approach

Data Sources

Estimation Methods

Model Evaluation

Covariate Analysis

Statistical Rigor Assessment
The hSIO model parameters were estimated by fitting to experimental time-course data for each drug, though the specific estimation algorithm is not detailed in the main text. The PBPK models were built in Simcyp V24 using standard platform methods. The clinical translation relies on a nonlinear multinomial logistic regression relating enterocyte-exposure-based villus injury to observed diarrhea frequencies; however, this was fitted to the dataset comprising all examined drugs rather than being externally validated. No formal diagnostics such as bootstrap confidence intervals, prediction errors, or cross-validation metrics are presented for the clinical predictions. The study acknowledges variation in reported clinical diarrhea incidence across trials but does not quantitatively account for this heterogeneity through meta-analytic or population modeling techniques. Sensitivity analyses around PBPK or epithelial model parameters are not reported.

⊠ Key Findings

The hSIO model quantified drug-specific proliferative cell killing and arrest for six oncology drugs (abemaciclib, ribociclib, poziotinib, lazertinib, mobocertinib, osimertinib), revealing markedly different toxicity profiles within drug classes. When the scale-free toxicity functions were integrated into the human epithelial dynamic model, enterocyte free drug concentration as a surrogate for crypt exposure predicted substantial epithelial injury for all diarrheagenic drugs tested, whereas plasma free drug exposure failed to predict injury for abemaciclib and mobocertinib. Predicted maximum villus injury ranged from approximately 3% for lazertinib to 100% for mobocertinib under enterocyte exposure scenarios, closely aligning with reported clinical diarrhea incidence. A fitted nonlinear multinomial logistic regression model translated predicted villus injury into probabilities of any-grade and grade ≥ 3 diarrhea that matched observed clinical frequencies across multiple trials and dosing regimens (Table 2), supporting the utility of hSIO-based QST for early safety differentiation and clinical dose selection.

💡 Clinical & Regulatory Implications

This QST framework supports early drug discovery decisions by enabling quantitative differentiation of GI toxicity liabilities among candidate molecules—exemplified by the clear separation of ribociclib and abemaciclib—before in vivo studies. For clinical development, it provides a rationale for using enterocyte-derived exposure rather than plasma exposure when establishing safety

margins for oral drugs, thereby reducing the risk of underestimating GI toxicity. The approach can inform first-in-human dose selection and schedule optimization by simulating virtual epithelial responses to proposed dosing regimens. From a regulatory perspective, the work aligns with the FDA Modernization Act 2.0 by demonstrating that advanced in vitro models combined with mechanistic mathematical modeling can reduce animal use while providing human-relevant safety predictions, though further validation across diverse compounds and populations will be needed for formal regulatory acceptance.

Strengths & Limitations

Strengths

- Novel multi-scale QST framework integrating human organoid pharmacology, PBPK local exposure prediction, and mechanistic epithelial dynamics to predict clinical toxicity.
- Use of scale-free toxicity functions enables direct translation from in vitro hSIOs to human epithelium independent of donor- or culture-specific proliferation rates.
- Demonstrates predictive utility across two mechanistically distinct drug classes (CDK4/6 inhibitors and EGFR-TKIs) with varying clinical diarrhea profiles.
- Human-relevant system circumventing cross-species translation limitations and supporting reduction of animal studies per FDA Modernization Act 2.0.
- Model code and MATLAB scripts publicly available through BioModels (MODEL2212120003), enhancing transparency and reproducibility.

Limitations (Acknowledged by Authors)

- hSIO experiments used a single donor; inter-donor variability was not experimentally assessed in this study.
- The quantitative link between predicted villus injury and diarrhea probability is phenomenological and requires validation with additional compounds.
- The framework is tailored to epithelial injury driven by antiproliferative effects and does not capture other diarrhea mechanisms such as secretory or immune-mediated GI perturbations.
- Reported clinical diarrhea incidence varies across trials for the same regimen, and the current framework does not incorporate population variability in pharmacokinetics or epithelial dynamics.
- PBPK enterocyte concentrations are used as a surrogate for crypt exposure, which remains physiologically unvalidated because crypt cells are not fully exposed to the lumen like enterocytes.

Limitations (Expert Review)

- No formal external validation was performed; the probabilistic diarrhea model was fitted to the same dataset of enterocyte exposure and clinical frequencies used to demonstrate predictive performance, limiting assessment of generalizability.
- Small number of drugs ($n = 6$) limits statistical confidence in the established exposure-toxicity-diarrhea relationship and may not capture the full spectrum of GI toxicities seen with antiproliferative agents.

- Uncertainty quantification (e.g., confidence or prediction intervals for predicted villus injury and diarrhea probabilities) is not presented, making it difficult to assess the precision of clinical predictions.
- The hSIO model assumes a single mechanism (proliferative cell killing/arrest) for all drugs, which may oversimplify agents with multi-modal toxicity such as EGFR-TKIs that also affect secretory pathways.
- No sensitivity analysis on key PBPK or epithelial model parameters was reported to understand drivers of prediction variability.

Generalizability

The findings are generalizable to oral antiproliferative oncology drugs where GI toxicity is driven by crypt proliferative cell impairment, but they may not extend to parenteral agents, drugs with predominant secretory or inflammatory diarrhea mechanisms, or compounds with significant active metabolite contributions not captured by the parent drug PBPK model. The enterocyte-exposure insight is broadly applicable to oral drug development, though the specific probabilistic model requires expansion across more drug classes and patient populations before regulatory acceptance.

Future Directions

Expert Commentary

Bottom Line

For practicing pharmacometricians, this paper provides a compelling case that plasma exposure alone is insufficient for predicting GI toxicity of oral antiproliferative drugs; local enterocyte free concentration must be used as the crypt exposure surrogate within a mechanistic translation framework. The integration of hSIO-derived toxicity parameters with PBPK and epithelial dynamics offers a actionable, human-relevant strategy for early candidate differentiation, first-in-human dose selection, and clinical trial design. Teams should consider adapting this QST approach —leveraging the publicly available code—for internal safety margin calculations, while planning for population variability and external validation to strengthen regulatory submissions.

Figures

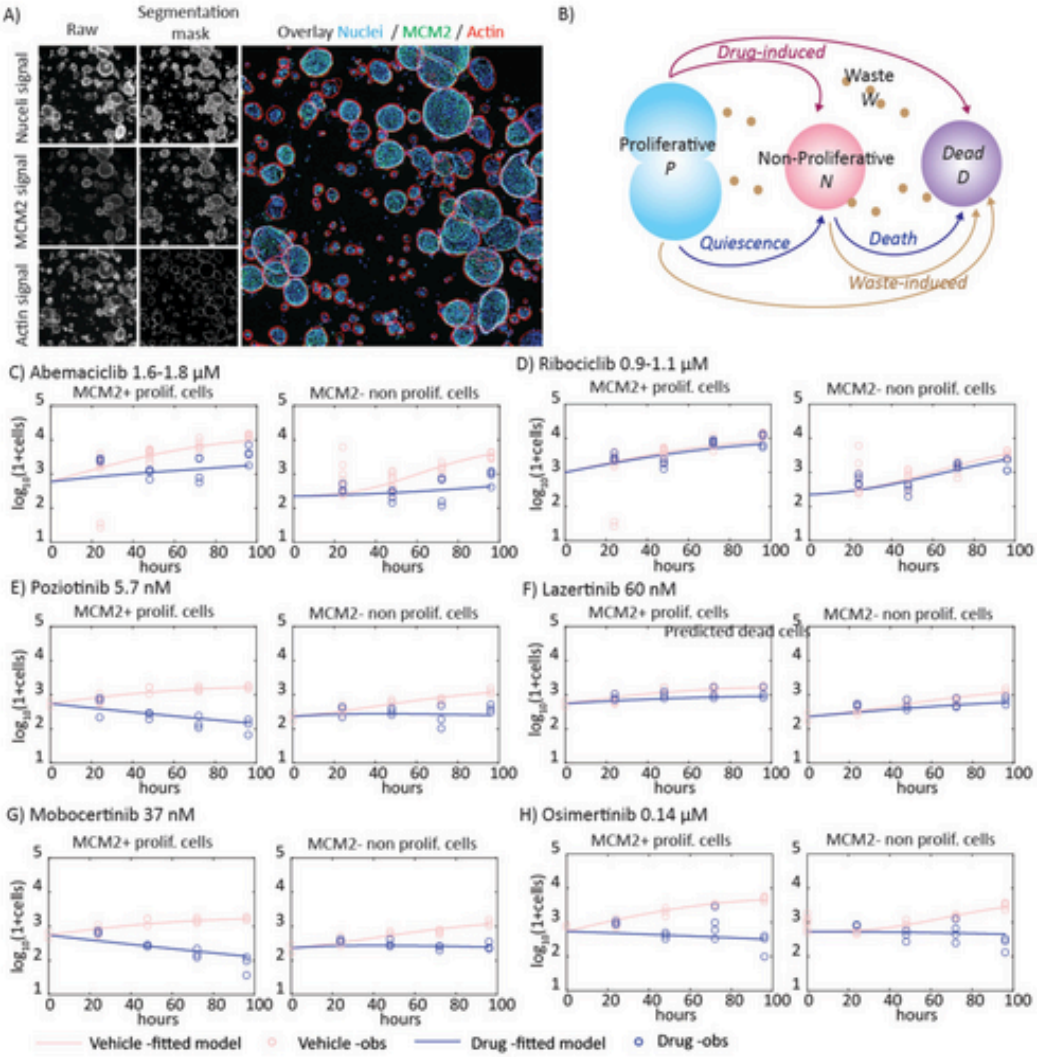

Figure 1: FIGURE 1: Modeling cell dynamics of drug-challenged hSIOs. (A) Example images (4×) of hSIOs stained for MCM2, Actin cytoskeleton and nuclei. (B) Schematic of our mathemati

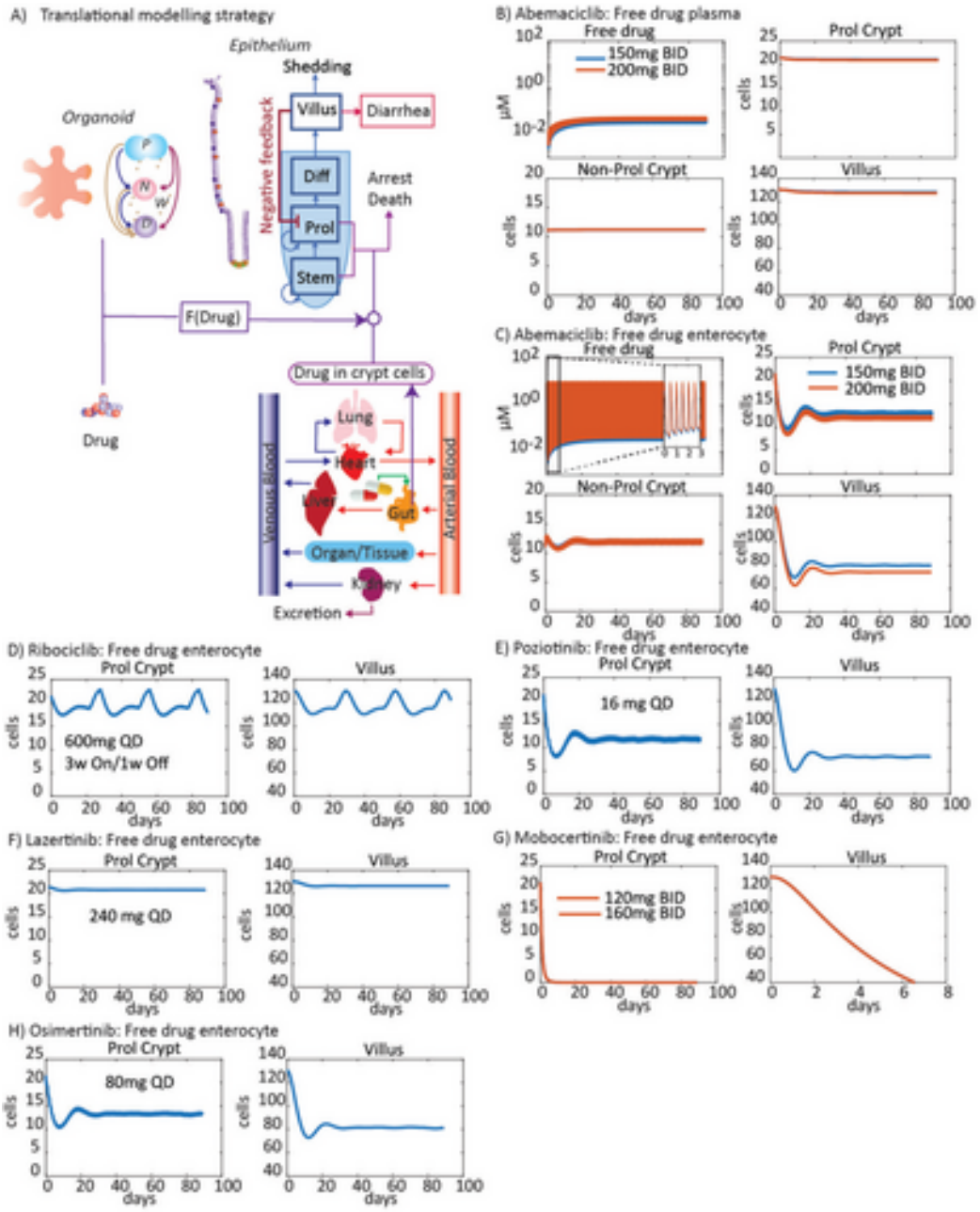

Figure 2: FIGURE 2: Translational approach from hSIO toxicity into epithelial injury. (A) Translational modeling approach from hSIOs to the human epithelium that quantifies the resp

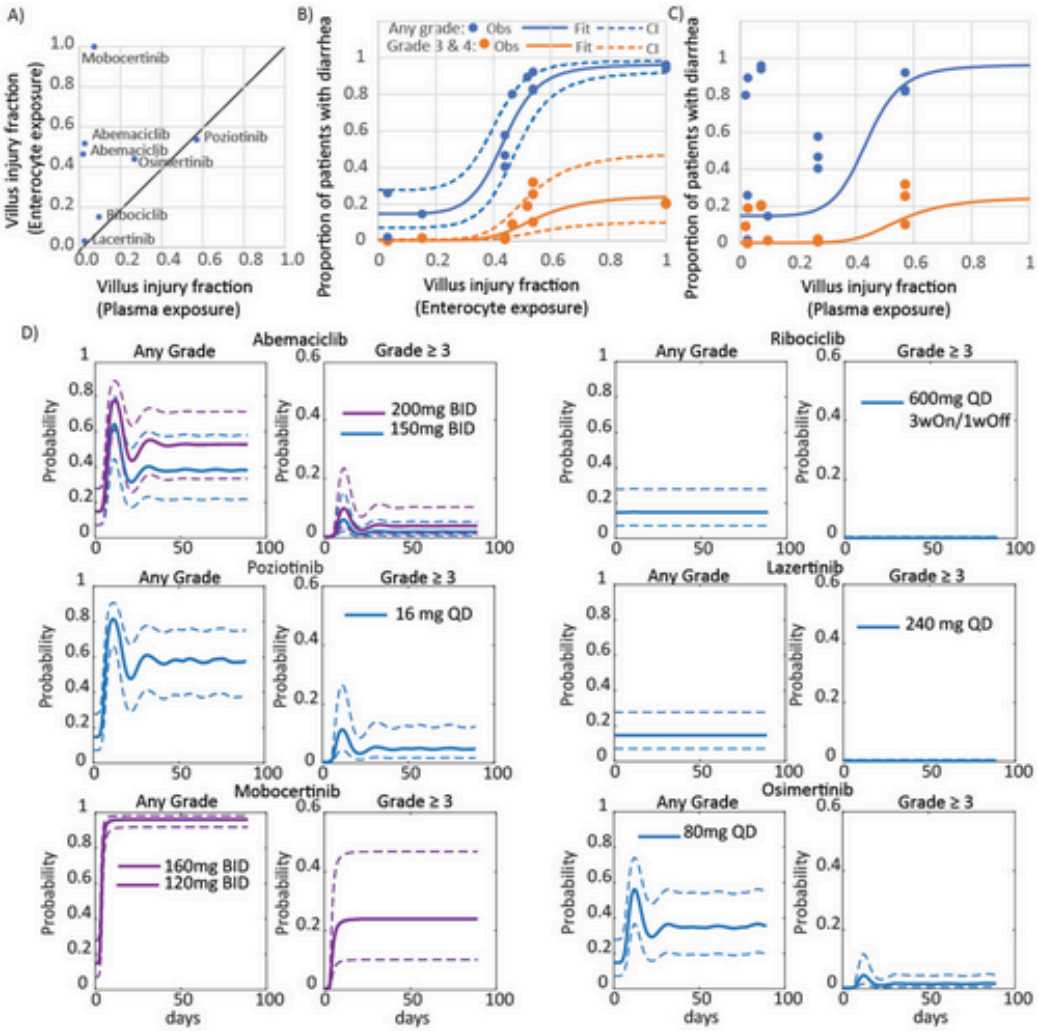

Figure 3: FIGURE 3: Predicting clinical incidence of diarrhea based on epithelial injury. (A) Comparison between the predicted maximum villus injury induced by plasma and enterocyte